Online Dependence-Aware Conformal Risk Allocation for Receding-Horizon Control

A Working Technical Note for Implementation and Theory Development

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1 Problem Statement

We want a receding-horizon controller whose uncertainty sets satisfy a trajectory-level guarantee of the form

$$\mathbb{P}\left(x_t \in \mathcal{E}_t \;\; \forall t = 1, \ldots, N\right) \geq 1 - \alpha_{\text{target}}. \tag{1}$$

Equivalently, defining violation events

$$A_t := \{x_t \notin \mathcal{E}_t\}, \qquad V_t := \mathbf{1}\{A_t\}, \tag{2}$$

we require

$$\mathbb{P}\left(\bigcup_{t=1}^{N} A_t\right) \leq \alpha_{\text{target}}. \tag{3}$$

The standard Bonferroni–Boole bound gives

$$\mathbb{P}\left(\bigcup_{t=1}^{N} A_t\right) \leq \sum_{t=1}^{N} \mathbb{P}(A_t) = \sum_{t=1}^{N} \alpha_t. \tag{4}$$

This is valid but can be extremely conservative for long horizons. The idea developed here is to exploit temporal dependence. If violations at adjacent times tend to occur together, then the pairwise overlaps

$$\omega_{s,t} := \mathbb{P}(A_s \cap A_t) \tag{5}$$

represent probability mass that Bonferroni double-counts. If we can learn certified lower bounds

$$\omega_{s,t}^{\text{safe}} \leq \omega_{s,t}, \tag{6}$$

we can subtract some of this double-counted mass while preserving a valid upper bound on the union probability.

Important technical correction. The full expression

$$\sum_{t} \alpha_t - \sum_{s<t} \omega_{s,t} \tag{7}$$

is generally *not* a valid upper bound on $\mathbb{P}(\cup_t A_t)$. It is the second-order inclusion–exclusion partial sum, which has the wrong sign for a general upper bound. A safe implementable correction is to subtract overlaps only on a suitable acyclic edge set, such as a maximum-weight spanning tree or forest. This is often called a Kounias–Hunter or Hunter-type pairwise Bonferroni upper bound. In this note, the phrase "Dawson–Sankoff-style" should be understood as a degree-two pairwise correction idea; for implementation safety, the graph-based upper bound below should be used unless a sharper theorem is explicitly proven.

2 Problem Setting

Consider a controlled stochastic system

$$x_{k+1} = f(x_k, u_k, w_k), \qquad x_k \in \mathbb{R}^n, \quad u_k \in \mathbb{R}^m, \tag{8}$$

operated in receding-horizon MPC. At MPC iteration k, the solver constructs a predicted nominal trajectory

$$\bar{x}_{0|k}, \dots, \bar{x}_{N|k}, \qquad u_{0|k}, \dots, u_{N-1|k}, \tag{9}$$

and an uncertainty set at each prediction index $t \in \{1, \dots, N\}$.

The intended uncertainty set is an ellipsoid

$$\mathcal{E}_{t|k}(c_{t|k}) := \left\{ x \in \mathbb{R}^n : (x - \bar{x}_{t|k})^\top A_{t|k}(c_{t|k})^{-1}(x - \bar{x}_{t|k}) \le 1 \right\}, \tag{10}$$

where

$$A_{t|k}(c_{t|k}) = c_{t|k}\, \bar{A}_{t|k}, \qquad c_{t|k} > 0. \tag{11}$$

Here $\bar{A}_{t|k} \succ 0$ is a base shape matrix determined by the nominal dynamics, covariance propagation, a learned world model, or another uncertainty propagation module, and $c_{t|k}$ is a scalar multiplier chosen by the controller or by a calibrated quantile map.

2.1 Why scalar multipliers matter

Learning dependence as a function of full matrices,

$$(A_{s|k}, A_{t|k}) \mapsto \omega_{s,t}, \tag{12}$$

is high-dimensional and sample inefficient. The scalar simplification replaces this with

$$\omega_{s,t} \approx \omega(c_s, c_t, \Delta t, \alpha_s, \alpha_t), \qquad \Delta t = t - s. \tag{13}$$

This is the key implementation simplification: the online dependence learner is tabular or low-dimensional rather than a high-dimensional function over S^n_{++}.

3 Marginal Calibration and the Ellipsoid Multiplier

For each horizon index t, the marginal risk is

$$\alpha_{t|k} := \mathbb{P}(x_{t|k} \notin \mathcal{E}_{t|k}). \tag{14}$$

The scalar multiplier $c_{t|k}$ should be chosen so that

$$\mathbb{P}(x_{t|k} \in \mathcal{E}_{t|k}(c_{t|k})) \ge 1 - \alpha_{t|k}. \tag{15}$$

In a conformal implementation, this is typically represented by a monotone quantile map

$$c_{t|k} = q_{t|k}(1 - \alpha_{t|k}), \tag{16}$$

where $q_{t|k}$ is obtained from calibration scores. Equivalently, because increasing c_t enlarges the ellipsoid, one may write

$$c_{t|k} = \psi_{t|k}(\alpha_{t|k}), \qquad \psi'_{t|k}(\alpha) < 0. \tag{17}$$

3.1 Gaussian ellipsoid probability option

If $x \sim \mathcal{N}(\mu, \Sigma)$, then

$$\mathbb{P}(x \in \mathcal{E}_A) = \mathbb{P}\left((x - \bar{x})^\top A^{-1}(x - \bar{x}) \le 1\right) \tag{18}$$

can be expressed as the CDF of a noncentral quadratic form after standardization. Let

$$z = \Sigma^{-1/2}(x - \mu), \qquad \tilde{\mu} = \Sigma^{-1/2}(\mu - \bar{x}), \qquad M = \Sigma^{1/2} A^{-1} \Sigma^{1/2}. \tag{19}$$

Then

$$(x - \bar{x})^\top A^{-1}(x - \bar{x}) = (z + \tilde{\mu})^\top M (z + \tilde{\mu}). \tag{20}$$

This can be evaluated numerically by Imhof or Davies-type methods, or approximated by moment matching. For an RTI controller, however, the conformal quantile map $c = \psi(\alpha)$ is often simpler and faster to use online.

4 Trajectory-Level Risk

At MPC iteration k, define the planned violation event

$$A_{t|k} := \{x_{k+t} \notin \mathcal{E}_{t|k}\}, \qquad V_{t|k} := \mathbf{1}\{A_{t|k}\}. \tag{21}$$

The desired joint condition is

$$\mathbb{P}\left(\bigcup_{t=1}^{N} A_{t|k}\right) \le \alpha^{\text{live}}_{\text{joint},k}. \tag{22}$$

Here $\alpha^{\text{live}}_{\text{joint},k}$ is an adaptive conformal risk budget updated online to target a desired long-run trajectory miscoverage rate.

The marginal risks are

$$\alpha_{t|k} = \mathbb{P}(A_{t|k}), \tag{23}$$

and the pairwise overlaps are

$$\omega_{s,t|k} = \mathbb{P}(A_{s|k} \cap A_{t|k}). \tag{24}$$

The corresponding binary Pearson correlation is

$$\rho_{s,t|k} = \frac{\omega_{s,t|k} - \alpha_{s|k}\alpha_{t|k}}{\sqrt{\alpha_{s|k}(1 - \alpha_{s|k})\alpha_{t|k}(1 - \alpha_{t|k})}}. \tag{25}$$

Although ρ is intuitive, ω is the quantity that directly appears in pairwise union bounds. Therefore, the recommended implementation is to learn ω directly and compute ρ only for diagnostics.

5 Safe Pairwise Bonferroni Correction

Let $A_1, \dots, A_N$ be arbitrary events. For any acyclic graph $G = (\{1, \dots, N\}, E)$, a valid pairwise Bonferroni upper bound is

$$\mathbb{P}\left(\bigcup_{t=1}^{N} A_t\right) \le \sum_{t=1}^{N} \mathbb{P}(A_t) - \sum_{(s,t)\in E} \mathbb{P}(A_s \cap A_t), \tag{26}$$

provided E is chosen as a forest in a way compatible with the classical graph-based Bonferroni/Kounias–Hunter construction. The tightest such bound for a fixed set of certified overlap weights is obtained by choosing a maximum-weight spanning tree or forest.

Remark 1 (Why not subtract all pairs?)**.** *The expression $\sum_t \mathbb{P}(A_t) - \sum_{s<t} \mathbb{P}(A_s \cap A_t)$ is generally not an upper bound. It can be smaller than the true union probability because triple and higher-order intersections are ignored. This is the most important technical nuance in this project. If the theory later proves that a richer Dawson–Sankoff-type expression is valid under additional structure, that expression can replace (26). Until then, (26) is the safer implementation target.*

5.1 Finite-lag truncation

For computational tractability, impose

$$\omega_{s,t}^{\text{safe}} = 0 \qquad \text{whenever } t - s > K. \tag{27}$$

Thus the candidate edge set is

$$E_K := \{(s,t) : 1 \le s < t \le N,\ t - s \le K\}. \tag{28}$$

The graph correction uses a maximum-weight forest over E_K, with weights $\omega_{s,t}^{\text{safe}}$. This gives $O(NK)$ candidate edges instead of $O(N^2)$.

5.2 Risk-allocation constraint

Let $F_k \subseteq E_K$ be the maximum-weight forest selected at MPC iteration k. The joint-risk constraint used by the optimizer is

$$\sum_{t=1}^{N} \alpha_{t|k} - \sum_{(s,t)\in F_k} \omega_{s,t|k}^{\text{safe}} \le \alpha_{\text{joint},k}^{\text{live}}. \tag{29}$$

If all $\omega_{s,t}^{\text{safe}} = 0$, this reduces to vanilla Bonferroni.

6 Learning Pairwise Overlap Online

6.1 What should be learned?

The recommended learned target is

$$\omega(c_s, c_t, \Delta t, \alpha_s, \alpha_t) := \mathbb{P}(V_s = 1, V_t = 1 \mid c_s, c_t, \Delta t, \alpha_s, \alpha_t). \tag{30}$$

This is preferable to learning ρ because the pairwise correction requires a lower bound on $\mathbb{P}(A_s \cap A_t)$, not merely a correlation estimate.

6.2 Discretized table

Discretize the input space into cells

$$j = \mathrm{bin}(c_s, c_t, \Delta t, \alpha_s, \alpha_t). \tag{31}$$

For each cell j, maintain

$$\mathsf{Table}[j] = \left(n_j, \hat{\omega}_j, \widehat{\mathrm{Var}}_j, M2_j\right), \tag{32}$$

where n_j is the number of completed observations assigned to the cell, $\hat{\omega}_j$ is the empirical mean of overlap indicators, and $M2_j$ is Welford's running second-moment accumulator, to keep track of and estimate quantities online.

For a completed sample, define

$$Y_j^k := V_{s|k} V_{t|k} \in \{0, 1\}. \tag{33}$$

Then update

$$n_j \leftarrow n_j + 1, \tag{34}$$

$$\hat{\omega}_j \leftarrow \hat{\omega}_j + \frac{Y_j^k - \hat{\omega}_j}{n_j}, \tag{35}$$

$$M2_j \leftarrow M2_j + (Y_j^k - \hat{\omega}_j^{\mathrm{old}})(Y_j^k - \hat{\omega}_j). \tag{36}$$

The empirical variance estimate is

$$\widehat{\mathrm{Var}}_j = \frac{M2_j}{\max\{1, n_j - 1\}}. \tag{37}$$

6.3 Lower-confidence correction

To avoid, over estimating the overlap values (which could result in below deisred coverage values). We can compute a *safe* overlap: The safe overlap used in the controller is

$$\omega_j^{\mathrm{safe}} = \max\{0, \hat{\omega}_j - M_j\}. \tag{38}$$

A simple fixed-time Hoeffding margin is

$$M_j^{\mathrm{H}}(n_j, \delta_j) = \sqrt{\frac{\log(1/\delta_j)}{2\max\{1, n_j\}}}. \tag{39}$$

A less conservative empirical Bernstein-style margin is

$$M_j^{\mathrm{EB}}(n_j, \delta_j) = \sqrt{\frac{2\widehat{\mathrm{Var}}_j \log(3/\delta_j)}{\max\{1, n_j\}}} + \frac{3\log(3/\delta_j)}{\max\{1, n_j\}}. \tag{40}$$

A time-uniform adaptive-Hoeffding-inspired margin can be used when table cells are queried at random stopping times:

$$M_j^{\mathrm{AH}}(n_j, \delta_j) = \sqrt{\frac{a\log(\log_c n_j + 1) + b_j}{\max\{1, n_j\}}}, \tag{41}$$

with recommended constants such as $a = 0.6$, $c = 1.1$, and b_j chosen so the desired cell-level failure probability is achieved. For example, using the corollary form

$$12e^{-1.8b_j} \le \delta_j \quad \Rightarrow \quad b_j \ge \frac{1}{1.8}\log\frac{12}{\delta_j}. \tag{42}$$

For a two-sided confidence sequence, replace δ_j by $\delta_j/2$.

6.4 Confidence allocation across cells

Let J denote the number of table cells that may be queried. A conservative choice is

$$\delta_j = \frac{\delta_{\text{conf}}}{J}. \tag{43}$$

If only a subset of cells is active, one can allocate confidence only over active cells, but this should be done carefully to avoid data-dependent undercoverage. For a first implementation, use the conservative uniform allocation.

7 Delayed History Buffer

The history buffer is necessary because MPC predictions are made before the corresponding future violations are observable.

At MPC iteration k, for every pair (s,t) with $1 \le s < t \le N$ and $t - s \le K$, store a record

$$r = \left(k, s, t, k+s, k+t, c_{s|k}, c_{t|k}, \alpha_{s|k}, \alpha_{t|k}, \mathcal{E}_{s|k}, \mathcal{E}_{t|k}\right). \tag{44}$$

The fields $k+s$ and $k+t$ are the absolute future times at which the two predicted events can be evaluated.

When the physical system reaches absolute time $k+t$, both x_{k+s} and x_{k+t} have been observed. Then compute

$$V_{s|k} = \mathbf{1}\{x_{k+s} \notin \mathcal{E}_{s|k}\}, \qquad V_{t|k} = \mathbf{1}\{x_{k+t} \notin \mathcal{E}_{t|k}\}. \tag{45}$$

The overlap sample is

$$Y^k_{s,t} = V_{s|k} V_{t|k}. \tag{46}$$

Map the *original planned* values

$$(c_{s|k}, c_{t|k}, t-s, \alpha_{s|k}, \alpha_{t|k}) \tag{47}$$

to a table cell and update that cell.

Remark 2 (Use original values)**.** *The values $c_{s|k}, c_{t|k}, \alpha_{s|k}, \alpha_{t|k}$ must be the values used when the ellipsoids were originally planned. Do not use the most recent live α values. Using recent values would mismatch the event geometry and bias the overlap estimate.*

8 Adaptive Joint Risk Budget

The joint budget $\alpha^{\text{live}}_{\text{joint},k}$ can be updated with an ACI/DtACI-style rule. After a predicted trajectory has fully resolved, define the trajectory violation indicator

$$E_k := \mathbf{1}\left\{\exists t \in \{1,\dots,N\} : x_{k+t} \notin \mathcal{E}_{t|k}\right\}. \tag{48}$$

Then update

$$\alpha^{\text{live}}_{\text{joint},k+1} = \text{clip}_{[\alpha_{\min},\alpha_{\max}]}\left(\alpha^{\text{live}}_{\text{joint},k} + \eta_k(\alpha_{\text{target}} - E_k)\right). \tag{49}$$

If the observed trajectory violates the tube, then $E_k = 1$, and for $\alpha_{\text{target}} < 1$ the update decreases $\alpha^{\text{live}}_{\text{joint}}$, making the next MPC problem more conservative. If no violation occurs, the budget increases, allowing less conservative future sets.

DtACI can replace (49) by maintaining a pool of learning rates or PI-style controllers. The dependence-learning table is independent of this update, except that the stored α values are part of the table key.

9 MPC/RTI Optimization Formulation

A natural nonlinear MPC formulation is

$$\min_{\boldsymbol{x},\boldsymbol{u},\boldsymbol{c},\boldsymbol{\alpha}} \quad \sum_{t=0}^{N-1} \ell(x_{t|k}, u_{t|k}) + \ell_f(x_{N|k}) + \lambda_c \sum_{t=1}^{N} r_c(c_{t|k}) \tag{50a}$$

$$\text{s.t.} \quad x_{0|k} = x_k, \tag{50b}$$

$$x_{t+1|k} = f(x_{t|k}, u_{t|k}), \qquad t = 0, \dots, N-1, \tag{50c}$$

$$u_{t|k} \in \mathcal{U}, \qquad t = 0, \dots, N-1, \tag{50d}$$

$$x_{t|k} \in \mathcal{X} \ominus \mathcal{E}_{t|k}(c_{t|k}), \qquad t = 1, \dots, N, \tag{50e}$$

$$c_{t|k} = \psi_{t|k}(\alpha_{t|k}), \qquad t = 1, \dots, N, \tag{50f}$$

$$\sum_{t=1}^{N} \alpha_{t|k} - \sum_{(s,t)\in F_k} \omega^{\text{safe}}_{s,t|k} \le \alpha^{\text{live}}_{\text{joint},k}, \tag{50g}$$

$$\alpha_{\min} \le \alpha_{t|k} \le \alpha_{\max}, \qquad t = 1, \dots, N. \tag{50h}$$

Here $r_c(c)$ penalizes uncertainty size. Examples include $r_c(c) = c$, $r_c(c) = c^2$, or $r_c(c) = \log\det(c\bar{A}_t)$.

9.1 RTI linearization

In an RTI scheme, solve one QP per control cycle. Use the previous solution as a warm start. The quantile map is linearized as

$$c_{t|k} \approx \bar{c}_{t|k} + \psi'_{t|k}(\bar{\alpha}_{t|k})(\alpha_{t|k} - \bar{\alpha}_{t|k}). \tag{51}$$

The pairwise safe overlaps can be handled in increasing levels of sophistication:

(a) **Frozen-overlap QP:** query $\omega^{\text{safe}}_{s,t}$ at the warm-start $(\bar{c}_s, \bar{c}_t, \bar{\alpha}_s, \bar{\alpha}_t)$, select the maximum-weight forest, and treat the overlap correction as a constant in the QP.

(b) **Piecewise-constant table QP:** allow α and c to move within their current bins, so the table lookup remains fixed.

(c) **Linearized-overlap QP:** estimate local slopes of $\omega(c_s, c_t, \alpha_s, \alpha_t, \Delta t)$ and linearize the correction. This is more complex and should be added only after the frozen version works.

For a first implementation, use option (a).

10 Algorithm

Algorithm 1 Online Dependence-Aware Conformal MPC

Inputs: horizon N, lag truncation K, target joint risk α_{target}, confidence budget δ_{conf}, bin edges for c and α, initial table Table, initial $\alpha_{\text{joint}}^{\text{live}}$
Initialize every table cell with $n_j = 0$, $\hat{\omega}_j = 0$, $M2_j = 0$
Initialize empty delayed history buffer Buffer
for MPC iterations $k = 0, 1, 2, \ldots$ **do**
    Observe current state x_k
    Build base ellipsoids $\bar{A}_{t|k}$ and nominal centers $\bar{x}_{t|k}$
    Query Table at warm-start $(\bar{c}_s, \bar{c}_t, \Delta t, \bar{\alpha}_s, \bar{\alpha}_t)$ for all $(s,t) \in E_K$
    Compute $\omega_{s,t}^{\text{safe}} = \max\{0, \hat{\omega}_j - M_j\}$
    Select maximum-weight forest $F_k \subseteq E_K$ using weights $\omega_{s,t}^{\text{safe}}$
    Solve the RTI QP with joint-risk constraint (29)
    Apply first control input $u_{0|k}$
    for all pairs (s,t) with $1 \le s < t \le N$, $t - s \le K$ **do**
        Push record $(k, s, t, k+s, k+t, c_{s|k}, c_{t|k}, \alpha_{s|k}, \alpha_{t|k}, \mathcal{E}_{s|k}, \mathcal{E}_{t|k})$ into Buffer
    end for
    Pop all records from Buffer whose absolute time $k + t$ has been observed
    for each popped record **do**
        Compute $V_{s|k}$ and $V_{t|k}$ using the stored ellipsoids
        Set $Y = V_{s|k} V_{t|k}$
        Map stored $(c_{s|k}, c_{t|k}, t - s, \alpha_{s|k}, \alpha_{t|k})$ to cell j
        Update Table$[j]$ using Y
    end for
    When a full trajectory-level prediction resolves, compute $E_k = \mathbf{1}\{\exists t : V_{t|k} = 1\}$
    Update $\alpha_{\text{joint}}^{\text{live}}$ using (49) or DtACI
end for

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